Sadia Afrose

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RESEARCH INTEREST

AI/ML for wireless communication, channel estimation and prediction for cellular and Wi-Fi systems, and building wireless sensing and on-device learning systems for next-generation networks.

Keywords: Wireless Systems, 5G NR, Sub-6 GHz, LTE, UMTS (WCDMA), Wi-Fi 7 (802.11be), 3GPP, O-RAN, PHY/MAC, MIMO, OFDM, Channel Estimation (CSI), Signal Processing, Integrated Sensing and Communication (ISAC), Digital Twin, Simulation Design, Test Automation, Systems Performance Analysis

EXPERIENCE

The Ohio State University Columbus, Ohio

Graduate Research Fellow, Cosine Lab, Advisor: Prof. Kannan Athreya Aug 2025 – Present

Physics-Guided ML for 5G NR Uplink Channel State Information (CSI) Prediction

- Built a 5G NR PHY measurement pipeline that captures uplink SRS-based CSI on srsRAN and OCUDU (O-RAN 7.2 split) with USRP SDRs and GNU Radio, and set up end-to-end LTE and 5G NR test networks with COTS UEs and SDR-based eNB/gNB/UE.
- Collected wideband 5G NR OFDM SRS/CSI datasets across indoor, lobby and rooftop environments on the POWDER testbed, and validated the CSI against 3GPP channel models (EPA/EVA/ETU) using Amarisoft UE simulator test equipment.
- Developing physics-guided ML, rather than black-box AI, for a generalizable model that predicts full-band CSI from partial-band SRS; trained only on a physics-based digital twin of synthetic channels (no real data) to learn physical channel parameters.

Wi-Fi 7 Wireless Sensing for Healthcare

- Building a Wi-Fi sensing system on HPE Aruba AP-755 next-generation Wi-Fi (Wi-Fi 7, 802.11be) access points, configuring the radios to collect CSI from commercial devices (mobile phones, laptops) on tri-band 2.4, 5 and 6 GHz at up to 160 MHz.
- Deploying the system in a hospital study for contactless seizure detection, and designing deep learning models for CSI-based sensing of human activity and gestures.

Over-the-Air Federated Learning on a Real 5G Cellular Setup

- Designing a multi-UE over-the-air computation (AirComp) PHY layer on a real sub-6 GHz 5G NR setup with SDRs and COTS UEs, enabling on-device learning where UEs train locally and aggregate updates over the air.
- Developing OFDM signal processing for synchronization and channel estimation that keeps independently clocked UEs phase-coherent without a shared clock, including under mobility.
- Building the simulation design (3GPP and ray-traced channels) and Python test automation for hardware experiments, evaluating airtime overhead and spectrum sharing with a live 5G cell.

Ericsson Network, Robi Axiata PLC Dhaka, Bangladesh

Senior Associate, Radio Access Network (RAN) Engineer, RAN Operations Aug 2023 – Jul 2025

- Performed cellular systems performance analysis and optimization of a live 2G (GSM), 3G (UMTS/WCDMA) and 4G LTE RAN, tracking all RAN KPIs, tuning power, frequency and bandwidth parameters, and reallocating unused hardware, power and bandwidth resources to enhance network performance.
- Deployed and integrated Massive MIMO systems and Ericsson Antenna Integrated Radios (AIR) to increase cell capacity and throughput, executing changes through Ericsson Network Manager (ENM) and WinFOIL.
- Built Python-based network analysis and automated shell scripts for routine RAN operations, and resolved complex radio network issues with vendors and cross-functional teams.

EDUCATION

The Ohio State University — Columbus, Ohio
Ph.D. in Computer Science and Engineering — Aug 2025 – Present

Bangladesh University of Engineering and Technology (BUET) — Dhaka, Bangladesh
B.Sc. in Electrical and Electronic Engineering — Mar 2018 – May 2023
Major: Communication and Signal Processing

PUBLICATIONS

- **Efficient Defense Against First Order Adversarial Attacks on Convolutional Neural Networks**
  S. Afrose, S. Karnine, H. Imtiaz · Artificial Intelligence and Applications, 2025
- **Aquatic Robot for Trash Removal and Water Quality Monitoring Built from Recycled Materials**
  S.K. Das, **S. Afrose**, S. Saha, R. Mashtura, F. Fardaous · ICECE 2024 · Best Paper Award

SKILLS

Standards	5G NR, LTE, UMTS (WCDMA), GSM, Wi-Fi 7 (802.11be), 3GPP, O-RAN (7.2 split, CU/DU, Near-RT RIC, xApps)
PHY & DSP	MIMO, Massive MIMO, OFDM, SRS/CSI, Bayesian (LMMSE) Channel Estimation, AirComp
Theory	Communication Theory, Digital Signal Processing, Random (Stochastic) Processes, Probability
AI/ML	Deep Learning, Federated Learning, PyTorch, TensorFlow/Keras, NumPy, Pandas
Programming	Python, C/C++, MATLAB, Shell Scripting, Linux, Embedded C, Assembly, Verilog HDL
Testbeds	Sionna RT, 3GPP TDL, USRP X310/B210, GNU Radio, srsRAN, OCUDU, Open5GS, POWDER
Test Tools	Amarisoft UE Simulator, Wireshark, Ericsson ENM, WinFOIL

SELECTED PROJECTS

- **Deep Learning for Micro-Doppler Radar Gesture Recognition** [Repository]: Applied vision transformers to Doppler-time radar spectrograms (DopNet) for RF sensing of hand gestures.
- **Intelligent Gesture-Based Security System** [Repository]: Designed a contactless embedded security system unlocked by a stored hand-gesture pattern, with gesture sensing and recognition simulated in MATLAB, VSPE, Arduino and Proteus.
- **IoT Remote Heart Rate Monitoring** [Repository]: Built an IoT health-monitoring system that processes ECG data on Arduino and MATLAB and reports to an IoT cloud with automated alerts.
- **Deep Learning for Time-Series Forecasting** [Repository]: Forecasted daily COVID-19 cases in Bangladesh with LSTM, MLP and Prophet models in Python, comparing their prediction performance.

AWARDS/HONORS

- Accepted, **Open6G+AI School** — 2026
  NSF-funded hands-on school on open-source AI-RAN and end-to-end 6G wireless networks, featuring workshops, tutorials, and hackathon-based training at Northeastern University
- Accepted, **CyberPowder Fellows Program** — 2026
  NSF-funded wireless networking hands-on training and research mentoring program led by University of Utah
- Recipient, **University Fellowship** — 2025-2026
  Awarded by The Ohio State University for exceptional academic qualification
- Recipient, **Best Paper Award** [Certificate] — 2024
  13th International Conference on Electrical and Computer Engineering (ICECE 2024)
- Recipient, **Dean's List Scholarship** — 2021-2023
  Academic Honor by BUET for attaining CGPA of 3.75 for two consecutive terms
- Recipient, **University Merit Scholarship** — 2021-2023
  Academic Honor by BUET to the top 10% students of a department in a semester
- **Ranked-3rd, Humanitarian Project Design Idea Contest** [Certificate] — 2022
  International Symposium on Social Implications of Sustainable Technology of IEEE Bangladesh Section Co-located Conferences
- Recipient, **Govt. Scholarship in Secondary School & Higher Secondary Certificate Examination** — 2015-2022
  Awarded by Ministry of Education, Bangladesh